Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 2 (Generated practice paper - Set 09)

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
3	
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6	
7	
8	
9	
10	
11	
TOTAL	

Do not write outside the box

Answer all questions in the spaces provided.

0 1 . 1 Which of the following would be most suitable for monitoring the movements of a whale over several months? Tick one box.

[1 mark]

A Radio collar ☐

B Acoustic / satellite tag ☐

C Pitfall trap ☐

D Geolocator tag recovered by recapture ☐

0 1 . 2 Explain why post-release monitoring is important in a captive breeding and release programme and describe **two** suitable methods.

[4 marks]

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0 2 Table 1 shows life-history features of three fish species and the age at which they reach a catchable size.

Table 1

Species	Age at catchable size / years	Age at first breeding / years	Eggs per female per year	Maximum lifespan / years
Atlantic cod	2	4	Millions	25
Orange roughy	20	30	Thousands	150
Herring	2	2	Tens of thousands	20

0 2 . 1 Use Table 1 to explain why orange roughy populations collapsed within ten years of large-scale fishing beginning in the late 1970s.

[4 marks]

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Do not write outside the box

0 2 . 2 Explain why herring populations recover from overfishing much faster than cod.

[2 marks]

0 2 . 3 Explain why the total mass of fish landed can stay the same while a population is being overfished.

[2 marks]

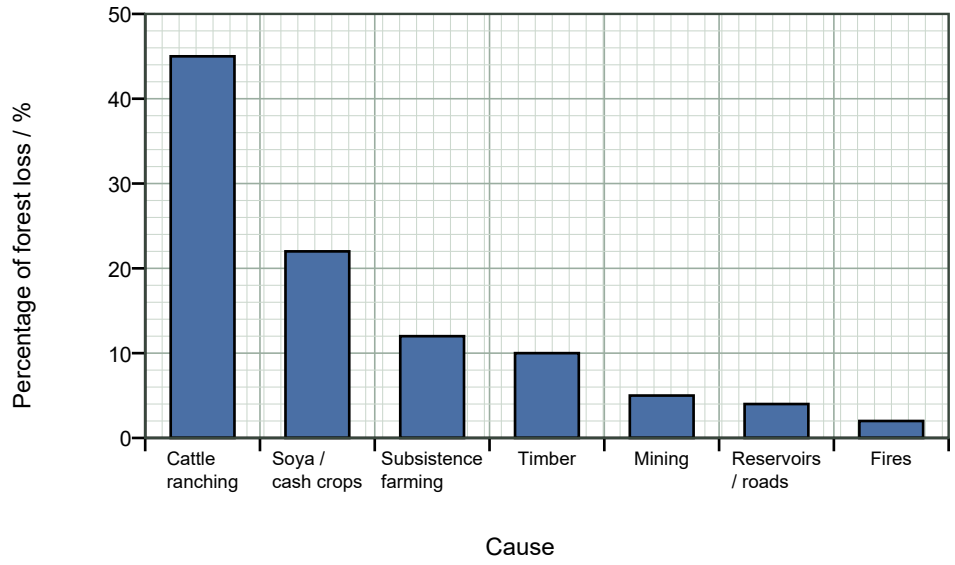
0 2 . 4 Describe how the Greenland shark and blue (common) skate illustrate the threat of by-catch to k-selected species.

[2 marks]

10

0 3 Figure 1 shows the main causes of tropical rainforest loss in a South American country between 2000 and 2020.

Figure 1



0 3 . 1 Explain why traditional 'slash and burn' subsistence agriculture was sustainable but has become a threat.

[3 marks]

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0 3 . 2 Explain why clearance for cattle ranching or cash crops is usually unsustainable.

[2 marks]

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0 3 . 3 Describe how each of the following threatens rainforest.

[3 marks]

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Mineral extraction .....

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HEP reservoirs .....

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Climate change .....

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0 3 . 4 Describe how a debt-for-nature swap works.

[2 marks]

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0 4 Table 1 shows the ecological footprint and biocapacity per person for four countries and the world in 2016.

Table 1

	Ecological footprint / gha per person	Biocapacity / gha per person	Human Development Index
USA	8.1	3.6	0.92
UK	4.4	1.4	0.92
China	3.6	1.0	0.75
Bangladesh	0.8	0.4	0.61
World	2.8	1.7	0.72

0 4 . 1 Define **ecological footprint** and **biocapacity**.

[2 marks]

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0 4 . 2 Explain what is meant by a **global hectare** and why it is used.

[2 marks]

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0 4 . 3 Use Table 1 to calculate how many Earths would be needed if everyone lived like the average US citizen.

[1 mark]

Answer ☐ Earths

0 4 . 4 Use Table 1 to describe the relationship between development and ecological footprint, and explain the concept of Earth Overshoot Day.

[3 marks]

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0 4 . 5 Suggest **two** ways a country such as the UK could reduce its footprint without reducing its HDI.

[2 marks]

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0 5 Table 1 shows the yield of a vegetable crop under different pest control methods over three years on one farm.

Table 1

Method	Mean yield / t ha^{-1}	Pesticide applications per year	Non-target insect species recorded in field margins
Chemical insecticide only	18.5	6	8
Crop rotation + beetle banks + hedgerows	16.2	0	31
Integrated control	18.1	1-2	26

0 5 . 1 Explain how crop rotation and beetle banks reduce pest damage.

[4 marks]

0 5 . 2 Use Table 1 to evaluate integrated control compared with the other two methods.

[3 marks]

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0 5 . 3 Describe how the sterile male technique controls insect pests and explain why it is species-specific.

[3 marks]

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0 6 . 1 Describe the resources obtained from forests. Give **one** example for each category.

[5 marks]

Timber .....

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Fuel .....

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Food .....

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Fibres .....

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Medicines .....

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0 6 . 2 Explain why forests are an important carbon reservoir and how they can be used to counteract climate change.

[3 marks]

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0 6 . 3 Explain how forests regulate the hydrological cycle and local climate.

[2 marks]

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0 7 . 1 Discuss the strategies that can be used to increase the sustainability of agriculture.

[9 marks]

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Extra space

0 7 . 2 State what is meant by **permaculture**.

[1 mark]

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0 8 . 1 Evaluate the extent to which aquaculture can replace fishing as a source of aquatic food.

[9 marks]

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Do not write outside the box

Extra space

0 8 . 2 State what is meant by a **fed** aquaculture species.

[1 mark]

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0 9 . 1 Explain **three** reasons why the development of new human activities has rarely been restricted by their environmental impacts.

[3 marks]

0 9 . 2 Explain why sedentary rural communities often exploit resources sustainably.

[2 marks]

0 9 . 3 Explain why urban populations are often unaware that the resources they use are being depleted.

[2 marks]

0 9 . 4 Explain why the growth of affluence has increased per capita environmental impacts.

[3 marks]

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1 0 . 1 Discuss the reasons why the conservation of biodiversity is important for human society.

[9 marks]

Do not write outside the box

Do not write outside the box

Extra space

1 0 . 2 State what is meant by **biodiversity**.

[1 mark]

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Write an essay on one of the following topics.

1 1 . 1 Discuss the extent to which agriculture can be made sustainable while feeding a growing human population.

[25 marks]

OR

1 1 . 2 Discuss the causes and consequences of deforestation and evaluate the extent to which forests can be exploited sustainably.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS